Arjuna JEE 2.0 2027

Maths

Basic Mathematics

DPP: 4

Total Duration - 53 Min

Q1 Given $\left(x^2+7x \geq 8\right)$ is called as ________.

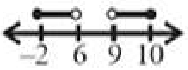
(A) Equation (B) Inequation
(C) Polynomial (D) None of these

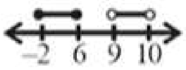
Q2 If we multiply whole inequality by a negative number then

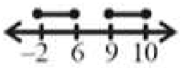
(A) sign of inequality remains same.
(B) sign of inequality gets reversed.
(C) depends, which negative number is multiplied.
(D) depends, both side of inequality are positive or negative.

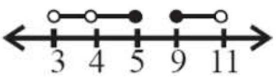
Q3 Correct number line represents for:
$x \in [-2,6) \cup (9,10]$

(A) 

(B) 

(C) 

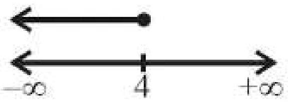
(D) 

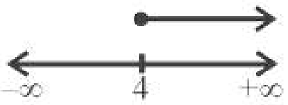
Q4 

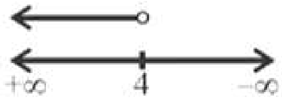
Correct option represents

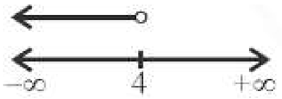
(A) $x \in (3,5] \cup [9,11)$
(B) x $\in (3,5] \cup [9,11)-\{4\}$
(C) x $\in (3,4) \cup (4,5) \cup [3,11)$
(D) x $\in (3,4) \cup (4,5] \cup (9,11)$

Q5 Correct number line representation for: $(x<4)$

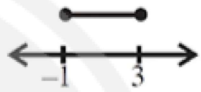
(A) 

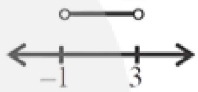
(B) 

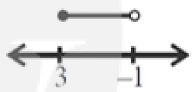
(C) 

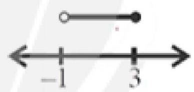
(D) 

Q6 Correct number line representation for:
$(-1<x \leq 3)$

(A) 

(B) 

(C) 

(D) 

Q7 Solution of inequality: $\frac{x-8}{2} \leq 5$

(A) $x<2$
(B) $x \geq 18$
(C) $x \leq 18$
(D) $x \geq 2$

Q8 Solution of inequality $(x-2)(x-3) \geq 0$

(A) $x \in (-\infty, 2) \cup (3, \infty)$
(B) $x \in (-\infty, 3)$
(C) $x \in (-\infty, 2] \cup [3, \infty)$
(D) $x \in [2,3]$

Q9 Solution of inequality $\frac{(x-1)(x-5)}{(x-3)}>0$

(A) $x \in (3,5)$
(B) $x \in (1,3) \cup (5, \infty)$
(C) $x \in (5, \infty)$

(D) $x \in (-\infty, 3)$

Q10 Solution of inequality $\frac{1}{x-5} < 3$ is

(A) $x \in (-\infty, 5)$
(B) $x \in \left(\frac{16}{3}, \infty\right)$
(C) $x \in \left(5, \frac{16}{3}\right)$
(D) $x \in (-\infty, 5) \cup \left(\frac{16}{3}, \infty\right)$

Q11 Values of $'x'$ satisfying: $\frac{x-1}{2-x} \geq 0$

(A) $x \in [1, 2]$
(B) $x \in [1, 2)$
(C) $x \in (-\infty, 1] \cup (2, \infty)$
(D) $x \in (1, 2)$

Q12 Solution of $\frac{x}{x+3} \leq 2$ is

(A) $x \in [-6, \infty)$
(B) $x \in (-\infty, -6]$
(C) $x \in (-\infty, -6] \cup (-3, \infty)$
(D) $x \in [-6, -3)$

Q13 Largest integral value of x satisfying:
$\frac{x^2-9}{(x-1)(x-6)} \leq 0$

(A) 5 (B) 3
(C) 4 (D) 6

Q14 Solution of inequality $-5 < x + 3 \leq 9$

(A) $x \in [-8, 6)$
(B) $x \in (-8, 6]$
(C) $x \in (-2, 6]$
(D) $x \in (-8, 6)$

Q15 Sum of all integral values of x satisfying:
$\frac{x+5}{1-x} \geq 0$

(A) -14 (B) $+14$
(C) -15 (D) 15

Q16 Solution of inequality $\left(x^2 - 12x + 35\right) < 0$

(A) $x \in (5, 7)$
(B) $x \in [5, 7)$
(C) $x \in (5, 7]$
(D) $x \in [5, 7]$

Q17 Values of $'x'$ satisfying : $\frac{x^2+x-2}{x^2-x-12} \geq 0$

(A) $x \in (-\infty, -3] \cup [-2, 1] \cup [4, \infty)$
(B) $x \in (-\infty, -3) \cup [-2, 1] \cup (4, \infty)$
(C) $x \in (-3, 4)$
(D) $x \in (-3, -2] \cup (4, \infty)$

Q18 Solution of $\frac{2x-x^2}{(3+x)(5-x)} < 0$ contains smallest negative integral value is:

(A) -3 (B) -2
(C) -1 (D) 0

Q19 Solution of $\frac{2x-3}{3x-5} \geq 3$ is:

(A) $x \in \left[\frac{5}{3}, \frac{12}{7}\right]$
(B) $x \in \left(\frac{5}{3}, \frac{12}{7}\right)$
(C) $x \in \left[\frac{5}{3}, \frac{12}{7}\right)$
(D) $x \in \left(\frac{5}{3}, \frac{12}{7}\right]$

Q20 Solution of $\frac{x^2-x-6}{x^2+6x} \geq 0$ is

(A) $x \in (-\infty, -6) \cup [2, 0) \cup [3, \infty)$
(B) $x \in (-\infty, -6) \cup [-2, 0) \cup [3, \infty)$
(C) $x \in (-\infty, -5) \cup [-2, 0) \cup [3, \infty)$
(D) $x \in (-\infty, -6) \cup [-2, 0) \cup [2, \infty)$

Q21 Solution of inequality: $2x^2 - 2x - 1 < 0$

(A) $x \in \left(\frac{1-\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$
(B) $x \in \left[\frac{1-\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right]$
(C) $x \in \left(-\infty, \frac{1-\sqrt{3}}{2}\right) \cup \left(\frac{1+\sqrt{3}}{2}, \infty\right)$
(D) $x \in (-1, 2)$

Q22 Solve the following inequalities

(i) $(x+1)(5x-1) > 0$
(ii) $x(x+3)(x-1) < 0$
(iii) $(2x-3)(x-2)(x-3) > 0$

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Answer Key

Q1	(B)
Q2	(B)
Q3	(B)
Q4	(B)
Q5	(D)
Q6	(D)
Q7	(C)
Q8	(C)
Q9	(B)
Q10	(D)
Q11	(B)
Q12	(C)
Q13	(A)
Q14	(B)
Q15	(C)
Q16	(A)
Q17	(B)
Q18	(B)
Q19	(D)
Q20	(B)
Q21	(A)
Q22	i) $x \in (-\infty, -1) \cup (1/5,\ \infty)$ ii) $x \in (-\infty, -3) \cup (0, 1)$ iii) $x \in (3/2, 2) \cup (3, \infty)$

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